10 CFR 50.73

CARY D. HARBOR
Vice President Nuclear
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102-08462-CDH/LMW
July 1, 2022

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Subject: **Palo Verde Nuclear Generating Station (PVNGS) Units 2 and 3**
Docket No. STN 50-529 and 50-530
License No. NPF-51 and NPF-74
Licensee Event Report 2022-001-00

Enclosed, please find Licensee Event Report (LER) 50-529/2022-001-00, that has been prepared and submitted pursuant to 10 CFR 50.73. This LER provides the cause and corrective actions for reported specified system actuations that occurred in Units 2 and 3 on May 4, 2022.

In accordance with 10 CFR 50.4, copies of this LER supplement are being forwarded to the Nuclear Regulatory Commission (NRC) Regional Office, NRC Region IV and the Senior Resident Inspector.

Arizona Public Service Company makes no commitments in this letter. If you have questions regarding this submittal, please contact Michael DiLorenzo, Department Leader, Regulatory Affairs, at (623) 393-3495.

Sincerely,

Katherine J Gil

Digitally signed by Gil, Katherine J(Z05492)
Reason: Signing for Cary Harbor per delegation of signature authority.
Date: 2022.07.01 14:44:17 -07'00'

CDH/LMW

Enclosure

cc:	S. A. Morris	NRC Region IV Regional Administrator
	S. P. Lingam	NRC NRR Project Manager for PVNGS
	L. N. Merker	NRC Senior Resident Inspector for PVNGS

NRC FORM 366 (08-2020) U.S. NUCLEAR REGULATORY COMMISSION

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 08/31/2023

LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
Palo Verde Nuclear Generating Station (PVNGS) Unit 2	05000529	1 OF 5

4. Title
Units 2 and Unit 3 Emergency Diesel Generator Actuation on Loss of Offsite Power to Class 4.16 kV Buses

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name: PVNGS Unit 3	Docket Number: 05000530
05	04	2022	2022	001	00	07	01	2022	Facility Name:	Docket Number: 05000

9. Operating Mode: 1 /1 **10. Power Level** 100/100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ **Other** (Specify here, in Abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include Area Code)
Michael DiLorenzo, Department Leader, Regulatory Affairs	623-393-3495

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
B	EA	XFMR	W120	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ **Yes** (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On May 4, 2022, at approximately 1955 Mountain Standard Time, a valid actuation of the emergency diesel generators (EDGs) for Palo Verde Nuclear Generating Station Unit 2 A-train and Unit 3 B-train occurred due to an undervoltage condition on their respective 4.16 kilovolt (kV) safety buses. Both EDGs started and loaded as designed, including associated train essential spray pond pumps and the Unit 3 B-train auxiliary feedwater pump.

The loss of power to the Unit 2 and Unit 3 safety buses and resulting component actuations were the result of startup transformer NAN-X01 tripping offline. Unit 2 and Unit 3 both entered Technical Specification (TS) Limiting Condition for Operation (LCO) 3.8.1, Condition A, for one required offsite circuit inoperable. In addition, both units also entered TS 3.8.9, Condition A, for the loss of the Class 1E bus, which was exited when the EDGs started and restored power to their respective Class 1E buses. LCO 3.8.1, Condition A, was exited by both units on May 5, 2022.

The direct cause of the undervoltage condition and the EDG and pump actuations was startup transformer NAN-X01 tripping offline. The direct cause of the event was attributed a grounded shield wire at the start-up transformer NAN-X01 W cabinet for the NAN-S06E cubicle differential relay current transformer (CT) from what should have been an unrelated billing and metering CT landed at a common terminal board location, resulting in circulating ground current.

No similar events have been reported by PVNGS in the last three years due to the same initial cause.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Palo Verde Nuclear Generating Station	05000-529	2022	001	00

NARRATIVE

All times are Mountain Standard Time and approximate unless otherwise indicated.

1. REPORTING REQUIREMENT(S):

This Licensee Event Report (LER) is being submitted pursuant to 10 CFR 50.73 (a)(2)(iv)(A) to report a valid automatic actuation of the circuitry that starts the emergency diesel generator (EDG) (EIIS: EK) for Unit 2 and Unit 3 following an undervoltage condition on the affected safety bus for each unit on May 4, 2022. The event included actuations of the associated essential spray pond (SP) pumps (EIIS: BS) and the Unit 3 B auxiliary feedwater (AF) pump (EIIS: BA). This event was reported to the NRC on May 5, 2022, via the event notification system (ENS 55878).

2. DESCRIPTION OF STRUCTURE(S), SYSTEM(S) AND COMPONENT(S):

The safety related equipment for each Palo Verde Nuclear Generating Station (PVNGS) unit is powered by one of two load groups (A-train and B-train). Either of the associated trains can provide power for safe plant shutdown. Each alternating current (AC) train includes one Class 1E 4.16 kilovolt (kV) bus (EIIS: EB).

The preferred and alternate power sources for each load group is offsite 525 kV AC power (EIIS: EK) and is supplied via the 13.8 kV secondary windings of two of the three startup transformers (EIIS: EA) to six 13.8 kV intermediate buses; Each unit receives 13.8 kV power from two of the intermediate buses. Class 1E 4.16 kV safety load group power is provided from the associated intermediate bus engineered safety feature (ESF) transformers. The standby power supply for each safety load group consists of one EDG (EIIS: EK), including its auxiliary and fuel systems. The standby power supply functions as a source of AC power for safe plant shutdown in the event of loss of preferred power and for post-accident operation of ESF loads.

3. INITIAL PLANT CONDITIONS:

On May 4, 2022, Palo Verde Units 2 and 3 were in Mode 1 (Power Operation) at 100 percent power, normal operating temperature and normal operating pressure. There were no inoperable structures, systems or components at the time that contributed to this event. Unit 1 was in Mode 5 (Cold Shutdown) for a refueling outage.

4. EVENT DESCRIPTION:

On May 4, 2022, at 1955, a valid loss of power (LOP) actuation occurred due to an undervoltage condition on the Unit 2 A-train and Unit 3 B-train safety buses which resulted in an automatic actuation of the circuitry that starts the Unit 2 A-train EDG and Unit 3 B-train EDG. The undervoltage condition on the Units 2 and 3 Class 1E 4.16 KV buses resulted from the startup transformer NAN-X01 tripping offline. Both EDGs started and loaded as designed. The EDG actuations were accompanied by the start of associated train essential SP pumps in both units. The Unit 3 B-train EDG actuation was also accompanied by the start of the Unit 3 B-train auxiliary feedwater pump, as required for the undervoltage condition. By design, actuation of the Unit 2 A-train steam-driven auxiliary feedwater pump was not required in response to the EDG actuation.

At 1955 on May 4, both Units 2 and 3 entered Technical Specification (TS) Limiting Condition for Operation (LCO) 3.8.1, Condition A for one required offsite circuit inoperable and LCO 3.8.9, Condition A for the loss of the Class 1E bus. Both units exited LCO 3.8.9, Condition A within one minute, after the Unit 2 A-train and Unit 3 B-train EDGs started and restored power to their respective Class 1E buses. Unit 2 exited LCO 3.8.1, Condition A on May 5 at 1525 after realigning the electrical path for offsite power to the A-train Class 1E bus from startup transformer NAN-X02. Unit 3 exited LCO 3.8.1, Condition A on May 5 at 1653 after realigning the electrical path for offsite power to the B-train Class 1E bus from startup transformer NAN-X03.

Unit 1 safety buses and required offsite circuits were unaffected by the NAN-X01 startup transformer trip. The redundant Unit 2 B-train and Unit 3 A-train safety buses were also unaffected by the event and remained energized by startup transformers NAN-X03 and NAN-X02, respectively.

Alarms in the Control Rooms indicated startup transformer NAN-X01 tripped offline, resulting in a loss of power to the Unit 2 A-train and Unit 3 B-train Class buses. Startup transformer NAN-X01 tripped from a differential relay (387) actuation. The differential relay tripped the 386T1 lockout relay. The lockout relay sent a trip signal to the other breaker being fed from startup transformer NAN-X01 as well as the switchyard relays. The switchyard relays tripped their lockout relays, which tripped the switchyard breakers feeding startup transformer NAN-X01. When startup transformer NAN-X01 tripped offline, the result was a loss of power on the Unit 2 A-train and Unit 3 B-train Class 1E buses. Due to the loss of power on the Class 1E buses, the Unit 2 A-train and Unit 3 B-train EDGs automatically started and restored power to their respective buses. All equipment responded as expected. Both of the units' associated SP pumps started in support of the actuations, as expected. In addition, the Unit 3 B-train AF pump automatically started, as expected. The Unit 3 B-train AF pump was not needed for steam generator level control and no AF valves repositioned. The Unit 3 B-train AF pump did not supply feedwater to the steam generators.

Unit 1 was in a refueling outage, and while the de-energization of startup transformer NAN-X01 did not impact the Unit 1 safety buses, there were impacts to Unit 1 at the same time as startup transformer NAN-X01 tripped offline. Simultaneous with the startup transformer NAN-X01 trip, Unit 1 received multiple alarms in the Control Room indicating a loss of power to three non-class load centers. Feeder breaker 13.8 kV 1E-NAN-S02E had tripped and locked out. The event tripped the 50G ground fault and the 50/51 overcurrent relays on 1E-NAN-S02E. At the same time, personnel in the field reported 480 V Load Center 1E-NGN-L02 exhibiting indications of an arcing fault. The feeder transformer for 1E-NGN-L02 had faulted to ground. Deposit material near the faulted phase of the transformer, which was submitted for analysis, was determined to be indicative of water residue. During a field walk down of load center 1E-NGN-L02, indications of prior water accumulation on the top of the load center were observed. Given the results of the deposited material analysis, visual inspection of the faulted load center transformer and the results of the field walk down, the likely cause of the fault on load center 1E-NGN-L02 was water intrusion. There were no active water leaks or identifiable possible sources of water observed for load center 1E-NGN-L02 during the walkdown.

The evaluation focused on why startup transformer NAN-X01 tripped during what should have been an unrelated ground fault on the Unit 1 load center 1E-NGN-L02. Initial troubleshooting determined the CT associated with the 387 differential relay for startup transformer NAN-X01 was wired so that a second ground path had been inadvertently introduced into the circuit. This mis-wiring caused circulating currents to develop in the circuit during the ground fault on the Unit 1 load center 1E-NGN-L02. This resulted in the startup transformer NAN-X01 tripping offline on a false 387 differential relay actuation, resulting in the subsequent loss of offsite power to one of the two safety buses in both Units 2 and 3. Though this mis-wiring became apparent during the load center 1E-NGN-L02 failure, other potential ground fault events at the site could have revealed this vulnerability.

5. ASSESSMENT OF SAFETY CONSEQUENCES:

There were no inoperable structures, systems or components at the time that contributed to this event. The EDGs responded as designed to the undervoltage condition on their respective safety buses. Essential SP and

AF pumps actuated as required for the LOP.

This event did not result in any challenges to the fission product barriers or result in the release of radioactive materials.

This event did not prevent the fulfillment of a safety function, nor did it result in a safety system functional failure as described by 10 CFR 50.73 (a)(2)(v).

6. CAUSE OF THE EVENT:

The direct cause of the undervoltage condition and the EDG and pump actuations was startup transformer NAN-X01 tripping offline. The direct cause of the event was attributed to a grounded shield wire at the start-up transformer NAN-X01 W cabinet for the NAN-S06E cubicle differential relay current transformer (CT) from what should have been an unrelated billing and metering CT landed at a common terminal board location, resulting in circulating ground current. Inadequate engineering rigor in developing the modification engineering documents for implementation and testing allowed the new CT wiring to be landed on a terminal connection that was already in use and failed to identify the additional ground during the design validation testing.

The following factors contributed to the event:

- Inadequate translation of the modification engineering documents into field work instructions
- Design change process requirements for correcting field deviations from the modification design lacked specificity for tracking changes and identifying transportability, i.e. similar mis-wiring was identified and corrected during Unit 2 modification installation, but not subsequently corrected in Units 1 and 3.

The event initiating ground fault on load center 1E-NGN-L02 was the result of the transformer shorting to ground, likely caused by water intrusion. However, many other ground fault events would have had the same end result since the fault simply revealed the vulnerability due to the mis-wired shield wire. A ground fault on load center 1E-NGN-L02 should have only resulted in the loss of three non-class load centers and impacted Unit 1 only.

7. CORRECTIVE ACTIONS:

Immediate corrective actions were taken to restore offsite power to Unit 2 A-train and Unit 3 B-train Class 1E buses. Alternate supply power was provided from startup transformers NAN-X02 and NAN-X03, respectively, through their associated intermediate buses to the Class buses. The 13.8 kV to 480 V transformer feeding load center NGN-L02 in Unit 1 was replaced.

Corrective actions to address the causes of the event included inspection of the other startup transformers to determine extent of condition and addressing any issues. Startup transformer NAN-X02 was found to be wired appropriately, with only a single ground on the CT circuit. The modification was implemented on February 27, 2018. Startup transformers NAN-X03 and NAN-X01 were both found to be incorrectly wired during a modification implemented on January 3, 2018 and December 21, 2018, respectively, having two grounds in the CT circuit. The spare startup transformer does not have the modification installed and if the modification was implemented in the future, it would be conducted under a new work order, using current behaviors and the Standard Design Process. The additional shield wire on startup transformer NAN-X01 was determinated and spared. Determination of the additional shield wire on startup transformer NAN-X03 is currently being planned and scheduled under an existing work order in accordance with normal station work control processes.

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Corrective actions were taken in 2019 and 2020 to address a similar (in nature) design change event and an identified trend of design change errors that were occurring at that time. In addition, the design modification process and procedure corrective actions completed from a previous evaluation were directed at capturing the bases and acceptance criteria in design modification documents and post modification testing. These prior corrective actions were robust and, therefore, limit the potential extent of condition for this event. Since these changes were implemented subsequent to the startup transformer modification work package, they will help prevent similar issues in the future.

Additional corrective actions taken as a result of this event included briefing personnel in the design modification process as well as expected behaviors and lessons learned. Lastly, electrical inspection procedures were revised to include details and requirements for water intrusion inspection. Work orders were generated to perform a walk down of the three units' turbine building electrical enclosures (load centers and motor control centers) to identify past or potential water intrusion issues and address identified concerns.

8. PREVIOUS SIMILAR EVENTS:

No EDG or ESF actuations resulting from mis-wiring in startup transformers due to inadequate engineering rigor in developing the modification engineering documents have been reported by Palo Verde.

However, there was a similar sequence of events with a different direct cause in which a startup transformer was de-energized, causing two units to experience a loss of power on a single Class 1E electrical bus, resulting in the automatic start of the associated EDGs, essential SP pumps and an AF pump. The most recent sequence of events as previously described was reported in LER 2021-001-01 for Palo Verde Unit 1. However, the events documented in this 2021 LER were initiated by inadvertent personnel electrical contact, so the root cause was different.

The corrective actions from the previous LER event would not have prevented the subject 2022 event.

While not the direct cause of this event, there was indication of water intrusion on the 1E-NGN-L02 load center that likely caused the initiating fault which exposed the design weakness leading to the event. There have been other water intrusion events at Palo Verde in the past, however, none directly caused a similar event.